

Econometric Reviews

ISSN: 0747-4938 (Print) 1532-4168 (Online) Journal homepage: www.tandfonline.com/journals/lecr20

Modelling the persistence of conditional variances

Robert F. Engle & Tim Bollerslev

To cite this article: Robert F. Engle & Tim Bollerslev (1986) Modelling the persistence of conditional variances, Econometric Reviews, 5:1, 1-50, DOI: 10.1080/07474938608800095

To link to this article: https://doi.org/10.1080/07474938608800095

Published online: 21 Mar 2007.

Submit your article to this journal

Article views: 1391

View related articles

Citing articles: 128 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=lecr20

ECONOMETRIC REVIEWS, 5(1), 1-50 (1986)

MODELLING THE PERSISTENCE OF CONDITIONAL VARIANCES

Robert F. Engle and Tim Bollerslev

Department of Economics, UCSD
La Jolla, CA 92093

and

Department of Economics, Northwestern University
Evanston, IL 60201

Key Words and Phrases: Autoregressive conditional heteroskedasticity; ARCH; GARCH; integration in variance; Student-t distribution; conditional kurtosis; time aggregation; non-linear conditional heteroskedasticity; asset pricing; exchange rate determination.

ABSTRACT

This paper will discuss the current research in building models of conditional variances using the Autoregressive Conditional Heteroskedastic (ARCH) and Generalized ARCH (GARCH) formulations. The discussion will be motivated by a simple asset pricing theory which is particularly appropriate for examining futures contracts with risk averse agents. A new class of models defined to be integrated in variance is then introduced. This new class of models includes the variance analogue of a unit root in the mean as a special case. The models are argued to be both theoretically important for the asset pricing models and empirically relevant. The conditional density is then generalized from a normal to a Student-t with unknown degrees of freedom. By estimating the degrees of freedom, implications about the conditional kurtosis of these models and time aggregated models can be drawn. A further generalization allows the conditional variance to be a non-linear function of the squared innovations. Throughout, empirical estimates of the logarithm of the exchange rate between the U.S. dollar and the Swiss franc are presented to illustrate the models.

Copyright © 1986 by Marcel Dekker, Inc. 0747-4938/86/0501-0001$3.50/0

TABLE OF CONTENTS

I. Introduction.....

II. The Simplest ARCH Model.....

III. Survey of Applications and Extensions.....

IV. Simple Asset Pricing Theories.....

V. Multistep Forecasting with the GARCH Model.....

VI. Integrated Variance Models.....

VII. Distribution of Multistep Forecast Errors and Time Aggregation.....

VIII. Non-Linear Conditional Heteroskedasticity.....

IX. Solving the Asset Pricing Equation.....

X. Conclusion.....

Acknowledgements.....

References.....

I. INTRODUCTION

The economic theory of behavior under uncertainty forms the basis for much of modern finance and monetary theory. An agent must make a decision based upon the distribution of a random variable some time in the future. In many rational expectations models it is assumed that only the mean of the conditional distribution affects the decision, however for more general utility functions and risk averse agents, a measure of dispersion will also be of primary importance. Conventional econometric methods have not been responsive to the need to develop quantitative measures of risk and uncertainty. This is particularly clear in time series

analysis where the very convenient assumption of linear covariance stationary models implies that most measures of uncertainty would remain constant over time.

This paper will discuss the current research in building models of conditional variances using the Autoregressive Conditional Heteroskedastic (ARCH) and Generalized ARCH (GARCH) formulations. The paper will motivate the discussion by introducing a simple asset pricing theory which is particularly appropriate for examining forward and futures contracts with risk averse agents. A new class of models defined to be integrated in variance is then introduced. This new class of models include the variance analogue of a unit root in the mean as a special case. The models are argued to be of both theoretical importance for the asset pricing models as well as empirically relevant. The conditional density is then generalized from a normal to a Student-t with unknown degrees of freedom. By estimating the degrees of freedom implications about the conditional kurtosis of these models and time aggregated models can be drawn. A further generalization allows the conditional heteroskedasticity to be a non-linear function of the squared innovations. Throughout, empirical estimates of the logarithm of the exchange rate between the U.S. dollar and the Swiss franc are presented to illustrate the models. The data are weekly observations on the Wednesday spot closing price and were generously provided by Frank Diebold. The same data set was examined in Diebold and Nerlove (1985).

II. THE SIMPLEST ARCH MODEL

To illustrate the unsatisfactory nature of time series and standard econometric models for modelling risk and uncertainty, consider the first order autoregressive model:

$$(1) \qquad y_t = \phi y_{t-1} + \epsilon_t , \quad \epsilon_t \text{ i.i.d.},$$

where

$$E(\epsilon_t) = 0, \quad V(\epsilon_t) = \sigma^2$$

and for stationarity $|\phi| < 1$. The expected value of y_{t+1} is simply zero but the conditional mean of y_{t+1} given the past information equals:[1]

$$E_t(y_{t+1}) = \phi y_t$$

which obviously is a random variable depending on the information set at time t. Similarly, the variance and the conditional variance can be expressed as:

$$V(y_{t+1}) = \sigma^2/(1-\phi^2)$$

$$V_t(y_{t+1}) = E_t[y_{t+1} - E_t(y_{t+1})]^2 = \sigma^2$$

Therefore, even though the conditional and unconditional variances are different, unless $\phi = 0$, they are both constants. By repeated substitution in (1) we get

---

[1] Of course, here and in what follows the conditional expectations are only defined almost surely. Without the loss of any essential results we shall simply ignore the technicalities arrising from this.

$$(2) \qquad y_{t+s} = \phi^s y_t + \sum_{i=1}^{s} \phi^{s-i} \epsilon_{t+i}$$

where the first term equals the conditional expectation of y_{t+s} at time t and the second term gives the corresponding forecast error. Thus, the conditional variance of y_t s steps ahead can be computed by

$$(3) \qquad V_t(y_{t+s}) = \sigma^2 \sum_{i=1}^{s} \phi^{2(s-i)}$$

so that although the conditional variance depends upon the forecast horizon it does not depend on the available information set or the forecast origin, t.

In more general models, the conditional variance s steps into the future would depend on the available information set and could therefore change over time. One might claim that the success of time series models is attributable to the use of the conditional mean for forecasting rather than the unconditional mean. Presumably similar gains are available for variances from using more sophisticated models of the conditional variances. The ARCH model introduced in Engle (1982) was designed to generalize the class of models used in time series analysis with exactly this point in mind. The conditional variance as well as the conditional mean were parameterized as functions of the available information set. While this allowed a vast increase in the potential models only a few of the parameterizations have been particularly useful. These are the ARCH(q) and the GARCH(p,q) models to be described below.

The simplest generalization of the AR(1) model given in (1) has an ARCH(1) model for the errors

$$y_t = \phi y_{t-1} + \epsilon_t$$

$$E_{t-1}(\epsilon_t) = 0 \tag{4}$$

$$V_{t-1}(\epsilon_t) = h_t = \omega + \alpha \epsilon_{t-1}^2$$

where $|\phi| < 1$, $\omega > 0$ and $\alpha \geq 0$.

The first order ARCH model described in (4) has several interesting properties. The disturbances are serially uncorrelated but not independent as they are related through second moments. If $\alpha < 1$, the unconditional variance of ϵ_t is given by:

$$V(\epsilon_t) = \sigma^2 = \omega (1-\alpha)^{-1} .$$

Using this notation the conditional variance of ϵ_t can therefore be rewritten as:

$$h_t - \sigma^2 = \alpha (\epsilon_{t-1}^2 - \sigma^2) \tag{5}$$

so that the conditional variance will be above the unconditional variance whenever the squared surprise last period exceeds its unconditional expectation. The marginal distribution of y_t will be symmetric if the conditional distribution of ϵ_t is symmetric. If ϵ_t is conditionally normal, the fourth unconditional moment of ϵ_t will exceed $3\sigma^4$ so that the marginal distribution of ϵ_t exhibits fatter tails than the normal.

It follows immediately from (4) that the conditional mean and the conditional variance of y_{t+1} evaluated at time t may be expressed as:

$$E_t(y_{t+1}) = \phi\, y_t$$

$$V_t(y_{t+1}) = h_{t+1} = \omega + \alpha(y_t - \phi\, y_{t-1})^2$$

so that both the conditional mean and the conditional variance of the one-step ahead forecast depend on the available information set. In particular the conditional variance is increased by either large positive or negative surprises in y_t.

Multi-step forecasts from the AR(1)-ARCH(1) model can be computed using the expression for y_{t+s} given in (2) leading to:

$$V_t(y_{t+s}) = \sum_{i=1}^{s} \phi^{2(s-i)}\, E_t(h_{t+s}) \ .$$

Provided $\alpha < 1$ and $i > 2$, it follows from (5) by the law of iterated expectations that:

$$E_t(h_{t+i}) = \sigma^2 + \alpha\, E_t(h_{t+i-1} - \sigma^2)$$

and the multi-step conditional variance equals:

$$V_t(y_{t+s}) = \sigma^2 \sum_{i=0}^{s-1} \phi^{2i} + \alpha^{s-1}(h_{t+1}-\sigma^2) \sum_{i=0}^{s-i} \phi^{2i}\, \alpha^{-i}. \tag{6}$$

Thus, the conditional variance is clearly not independent of the current information set, but for long term forecasts and α less

than one the dependence on $(h_{t+1} - \sigma^2)$ becomes negligible and the expression in (6) is well approximated by (3).

Assuming ϵ_t to be conditionally normal distributed, it follows by the prediction error decomposition, that for a sample of T observations the log likelihood function is, apart from some initial conditions, given by

$$(7) \qquad \text{LogL} = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}(\log(h_t) + \epsilon_t^2 h_t^{-1}) \ .$$

Expressions for the gradient, the Hessian and Fishers information matrix are given in Engle (1982), along with a discussion of methods for maximization of (7).

All of the above arguments and calculations extend in a straight forward manner to the more general ARCH(q) model in which h_t depends linearly on q lagged values of ϵ_t^2,

$$V_{t-1}(\epsilon_t) = h_t = \omega + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2$$

where now $\alpha_i \geq 0$, $i = 1,\ldots,q$. For a more detailed discussion of the ARCH(q) model the reader is referred to the original paper by Engle (1982).

III. SURVEY OF APPLICATIONS AND EXTENSIONS

In a series of papers, the ARCH model has been analyzed, generalized, extended to the multivariate context, and used to

test for time varying risk premia in the term structure of interest rates and in other financial markets. These papers include Engle (1983) and Engle and Kraft (1983) where a measure of the variance of inflation are given. The ARCH model is extended to a multivariate framework in Kraft and Engle (1982). In Engle, Granger and Kraft (1984) and Granger, Robins and Engle (1984) bivariate ARCH models of inflation with changing covariances as well as variances are constructed. Engle, Brown and Stern (1984) and Granger and Engle (1984) examine the effectiveness of ARCH models for forecasting purposes. The power of ARCH tests and the finite sample properties of various ARCH estimators, are analyzed in Engle, Hendry and Trumble (1985) by means of Monte Carlo methods. In Bollerslev (1985a, 1985b) the Generalized ARCH or GARCH model is developed, and the GARCH model with conditionally Student-t distributed errors is studied in Bollerslev (1985c). Engle, Lilien and Robins (1985) and Bollerslev, Engle and Wooldridge (1985) examine the term structure of interest rates and a three asset Capital Asset Pricing Model (CAPM) to determine whether risk premia are varying over time. These papers introduce the ARCH in mean or ARCH-M model in a univariate and multivariate context respectively. Engle and Watson (1985) recast the GARCH-M model in a full information state space form.

In addition to this work, a variety of papers have begun to appear from different parts of the world. Without attempting an exhaustive list of these, particularly interesting are papers by Milhøj (1985a) who develops far more general moment conditions

than those in the original Engle (1982) paper. Linnel-Nemec (1984a, 1984b) establishes stationarity and ergodicity of ARCH models. Pantula (1984) and Weiss (1982) derive the limiting distribution of ARCH estimators in more general contexts. Pagan, Hall and Trivedi (1983), Weiss (1984) and Coulson and Robins (1985) provide empirical time series examples of ARCH and related models of changing variances. Domowitz and Hakkio (1985), Diebold and Pauly (1985), Diebold and Nerlove (1985), Milhøj (1985b), Hsieh (1985) and McCurdy and Morgan (1985), apply the ARCH, the ARCH-M and the GARCH model to the foreign exchange market. Amsler (1984a, 1984b) investigate whether using the risk premia estimated by ARCH models will make long bonds satisfiy the Shiller variance bounds. Poterba and Summers (1984) derive a pricing formula for stock market prices in the spirit of the asset pricing formulas presented in this paper. The price is related to its own variance, which is modelled as a simple AR(1) process. Similar ideas are employed in the paper by French, Schwert and Stambaugh (1985). Blanchard and Watson (1984) and Bodie, Kane and McDonald (1983, 1984) present evidence that macroeconomic and financial time series models can usefully be reformulated as a form of multivariate ARCH processes.

These applications can all be viewed in a common econometric framework with a focus on modelling the behavior of economic agents to risk in a time series context. The general model is a simultaneous equations system where the elements of the ARCH disturbance covariance matrix also appear as regressors. To establish notation, let y_t be a vector of endogenous random variables

and ψ_t the information set available at time t. Also, let μ_t and H_t denote the conditional mean and the conditional variance of y_t respectively:

$$\mu_t = E[y_t|\psi_{t-1}]$$

$$H_t = E[(y_t-\mu_t)(y_t-\mu_t)'|\psi_{t-1}] \ .$$

Finally, let X_t be a vector of weakly exogenous and lagged dependent variables in the sense of Engle, Hendry and Richard (1983). The conditional mean and variance of the general model can then be parameterized as:

$$\mu_t = X_t\beta + \text{vech}(H_t)\ \delta$$

$$H_t = A + (I \otimes X_t)'C\ (I \otimes X_t)$$

where $\text{vech}(H_t)$ denotes the vector of all the unique elements of H_t obtained by stacking the lower triangle of H_t, A is a positive definite matrix and C is a positive semi-definite matrix. This class of models represents the reduced form of a structural simultaneous equations system where the mean of the observed behavior is affected by the covariance matrix of all observed endogenous variables. Such a system could be described as:

$$B\ y_t + D\ \mu_t + F\ \text{vech}(H_t) + G\ X_t = \epsilon_t$$

which is closely related to Wallis (1980)'s rational expectations version of a simultaneous system except that in this case both conditional means and conditional variances and covariances enter the structural relation.

If in the above reduced form $\delta=0$, this is simply a multivariate heteroskedasticity model. If in addition X_t enters the heteroskedasticity function only through squares and cross products of $\epsilon_t = y_t-\mu_t$, the multivariate ARCH model of Kraft and Engle (1982) is obtained. The multivariate GARCH model of Bollerslev, Engle and Wooldridge (1985) has H_t depending on past squares of ϵ_t's and past values of H_t's. In effect, this corresponds to a different lag structure on past squared ϵ_t's.

In the univariate context with $\delta \neq 0$, the ARCH-M model is obtained. Engle, Lilien and Robins (1985) uses this type of model behavior in modelling the term structure of interest rates where the variance of the excess holding yield on a long bond is a determinant of the expected return on that asset. The ARCH-M model is extended to a multivariate setting in Bollerslev, Engle and Wooldridge (1985), where a Capital Asset Pricing Model (CAPM) with time varying betas is estimated. The expected return on an asset is proportional to the covariance with the market return, that is the CAPM gives the restriction

$$\mu_t = \delta H_t w_t$$

where w_t is the vector of value weights for all the assets, assumed to be in the information set. The results for a model with three risky assets: bills, bonds and stocks, are promising. There is strong evidence for a time varying conditional covariance matrix for the asset returns. The coefficient of aggregate relative risk aversion, δ, is statistically significant. The trivariate

model in which the risk premium is associated with the covariance with the market outperforms a set of univariate models where the premia depend only on the own variances. The model, however, does fail some of the more demanding specification tests suggesting further work along these lines might be fruitful.

IV. SIMPLE ASSET PRICING THEORIES

To motivate the new class of models called integrated in variance to be defined below, a very simple illustrative model of asset pricing will be presented. It has a familiar form and can be solved to give previously unreported pricing equations.

Models of asset pricing involve agents maximizing expected utility over uncertain future events. The competitive outcome sets a price for the asset today which equates supply with demand. For example suppose a representative agent must allocate his wealth, W_t, between shares of a risky asset q_t at a price p_t and those of a sure asset x_t with a price equal to 1. The shares of the risky asset will be worth y_{t+1} each at the end of the period, and the sure asset will be worth $r_t x_t$, where r_t denotes one plus the risk free rate. Note, that if there are no dividends $y_{t+1} = p_{t+1}$. Also, suppose the representative agent has a mean variance utility function in end of period wealth, $q_t y_{t+1} + r_t x_t$. With these assumptions the allocation problem becomes:

$$(8) \qquad \underset{q_t}{\text{Max}} \quad 2E_t(q_t y_{t+1} + r_t x_t) - \gamma_t V_t(q_t y_{t+1})$$
$$\text{s.t. } W_t = x_t + p_t q_t$$

which has the solution

$$(9) \qquad p_t = \frac{1}{r_t} E_t(y_{t+1}) - \frac{\gamma_t q_t}{r_t} V_t(y_{t+1}) \ .$$

If the outstanding stock of the risky asset is fixed at q, then (9) describes the asset pricing model. Typically, W_t, r_t and γ_t are also taken as constants in such a formulation, even though this is not required by the model. However, for ease of exposition we shall treat W, q, γ and r as constants in what follows.

It is worth considering exactly what is implied by the relation given in (9). Both the conditional mean and the conditional variance are functions of the conditioning information set which includes p_t. The situation is therefore similar to the Walrasian auctioneer who must declare a price known to all agents that clears the market. The equation therefore has no error term unless there are other sources of uncertainty and these may be problematic from a theoretical point of view. We are indebted to Roman Frydman for this observation.

If now the risky asset is interpreted as a forward contract for delivery in s periods, the price a pure speculator characterized by (8) would be willing to pay is simply

$$(10) \qquad p_t = r^{-s}[E_t(y_{t+s}) - \delta V_t(y_{t+s})]$$

where r^{-s} gives the discounted present value at the sure rate r and $\delta = \gamma q$. Note, p_t is the price paid in full at time t for delivery at time t+s. Of course, $p_{t+s} = y_{t+s}$ by definition. In

the foreign exchange market this model relates the price of a forward contract for delivery in s periods to the expected future value of the spot exchange rate y_{t+s} and the variance of that rate discounted at the sure domestic interest rate r. Typically it is assumed that $\delta=0$ so that there is no risk premium and the forward rate is an unbiased predictor of the future spot rate corrected for interest rate differentials.

A simple redating of the model shows that the price of the forward contract at time t+1 for $s \geq 2$ periods remaining to maturity can be expressed as:

$$p_{t+1} = r^{1-s} \left[E_{t+1}(y_{t+s}) - \delta V_{t+1}(y_{t+s})\right] .$$

Taking expectations at time t, multiplying by r^{-1} and subtracting from (10) gives:

$$p_t = r^{-1} E_t(p_{t+1}) - \delta r^{-s}\left[V_t(y_{t+s}) - E_t(V_{t+1}(y_{t+s}))\right] . \tag{11}$$

Now, suppose y_t can be represented by an infinite moving average process where the increments are uncorrelated but have time varying conditionnal variance h_t.

$$y_t = \sum_{i=1}^{\infty} \theta_i \epsilon_{t-i} = \theta(L)\epsilon_t$$

$$V_t(y_{t+1}) = V_t(\epsilon_{t+1}) = h_{t+1} . \tag{12}$$

It follows that

$$(13) \qquad V_t(y_{t+s}) = E_t\Big(\sum_{i=1}^{s} \theta_{s-i}\, \epsilon_{t+i}\Big)^2 = \sum_{i=1}^{s} \theta_{s-i}^2\, E_t(h_{t+i}) \ .$$

Consequently the expression in square brackets in (11) is simply $\theta_{s-1}^2 h_{t+1}$ leading to:

$$(14) \qquad p_t = r^{-1} E_t(p_{t+1}) - \delta r^{-s} \theta_{s-1}^2 \, h_{t+1}$$

which is the familiar formula for a one period holding yield with the explicit calculation of the effect of the changing variance of y_{t+s} for a risk averse agent. Notice, however, that the expectation of the final spot price no longer enters this equation directly, but only through its effect on $E_t(p_{t+1})$. Also, if θ_{s-1} becomes small for large s, the only relevant information about the risk of the contract is already incorporated in that expectation.

In the simple model above the only source of uncertainty derives from the future spot price to which the contract relates. In many other situations however, there is a flow of uncertain distributions which accrue to the owner of the asset. For example, the price of a share of a stock is presumably determined by the present discounted value of the expected dividend stream. The rate of interest on a long bond is conjectured by the expectations hypothesis to be the present discounted value of the sequence of short term rates expected during the life of the bond. The value of a fruit tree is the present discounted value of the expected

value of the fruit. In each case, the model is based on the assumption of risk neutral investors and exogenously given consumption.

However, risk averse investors would everything else equal pay less for streams of income the more uncertain the streams. The precise fashion in which the variability of future payoffs enters the asset pricing formulation depends among other things upon the utility function of the agents and the intertemporal substitutability of the payouts. The two simplest formulations are the variance of the discounted payoffs and the discount of the variance of the payoffs, i.e. either of the following two forms:

$$(15) \qquad p_t = \sum_{s=1}^{\infty} r^{-s}[E_t(y_{t+s}) - \delta\, V_t(y_{t+s})]$$

or

$$p_t = \sum_{s=1}^{\infty} r^{-s}E_t(y_{t+s}) - \delta\, V_t\Big(\sum_{s=1}^{\infty} r^{-s}y_{t+s}\Big)$$

where y_{t+1}, y_{t+2}, ... refer to the future income stream generated by the asset. The latter of these two can also be expressed as

$$(16) \qquad p_t = \sum_{s=1}^{\infty} r^{-s}E_t(y_{t+s}) - \delta \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} \omega_{ij}\mathrm{Cov}(y_{t+i}, y_{t+j}).$$

Under the same set of assumptions as in (12) about the process generating y_t (15) can be converted to a holding yield expression:

$$(17) \qquad p_t = r^{-1} \left[E_t(p_{t+1}) + E_t(y_{t+1}) - \delta\lambda\, h_{t+1} \right]$$

where λ depends upon the polynomial $\theta(L)$ and the risk free rate r. Assuming h_{t+1} is generated by an integrated GARCH(1,1) model, to be defined below, a similar expression holds for (16). The important feature of this result is that $V_t(p_{t+1})$ does not independently enter the asset pricing equation. The variance of next periods payout is sufficient information for risk averse agents. This differs from the Summers and Poterba (1984) formulation where $V_t(p_{t+1})$ enters the pricing equation explicitly.

Examining expressions (11) through (17) it is clear that if $\delta \neq 0$, the conditional variance of y_t in the future will affect the price of the asset today. The extent of the effect cannot be seen directly from (14) or (17) which are stated in terms of future expectations about the price, but can be seen from the discount expressions (10), (15) or (16). If variances can be forecast as in the ARCH example in the preceeding section, then the current information on y_t and the current conditional variance will have an effect on the current price. However, the size of the effect will depend upon the persistence of the variance. That is, it will depend upon how important current information is in predicting future variances. This question is of great importance since it explains how new information influences current asset prices.

V. MULTI-STEP FORECASTING WITH THE GARCH MODEL

Bollerslev (1985a) has recently introduced the Generalized ARCH or GARCH model which incorporates in a simple form many of the properties which have been found useful in ARCH estimation. A random variable ϵ_t is said to follow a GARCH(p,q) process if

$$E_{t-1}(\epsilon_t) = 0$$

$$V_{t-1}(\epsilon_t) = h_t = \omega + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i} \tag{18}$$

$$= \omega + \alpha(L)\epsilon_t^2 + \beta(L)h_t$$

where $\omega > 0$, $\alpha_i \geq 0$ and $\beta_i \geq 0$ for all i. This formulation generalizes the ARCH(q) model by allowing non-zero β_i's. Simple substitution reveals that the GARCH model is simply an infinite order ARCH model with exponentially decaying weights for larger lags. Thus, a low order GARCH model may have properties similar to high order ARCH models without the problems of estimating many parameters subject to non-negativity constraints. In fact, the ARCH models estimated in Engle (1983), Engle and Kraft (1983), and Engle, Lilien and Robins (1985), impose linearly declining weights in the α_i's so that the only free parameters are q and the sum of the weights. Selection of q can be addressed by model selection techniques but is somewhat awkward. Thus the GARCH model appears to be a natural and simple generalization of the ARCH model, and empirical evidence suggests that it fits as well or even better

than the ARCH model with linearly declining weights with roughly the same mean lag, see Bollerslev (1985a) for more details.

Sastry Pantula has pointed out that the GARCH model can be rewritten in an alternative and from a theoretical time series point of view maybe more interpretable form. Let $v_t \equiv \epsilon_t^2 - h_t$, then the conditional variance in expression (18) may be rearranged as:

$$(19) \qquad \epsilon_t^2 = \omega + \sum_{i=1}^{m} (\alpha_i + \beta_i)\epsilon_{t-i}^2 - \sum_{i=1}^{p} \beta_i v_{t-i} + v_t$$

where $m = \max\{p,q\}$, $\alpha_i \equiv 0$ for $i > q$ and $\beta_i \equiv 0$ for $i > p$. This reveals that ϵ_t^2 follows an ARMA(m,p) process with serially uncorrelated innovations v_t. Note, however, the peculiar feature that the innovations v_t have different support in each time period and consequently are heteroskedastic with a very complicated distribution. It is clear that the autoregressive part of the process is given by the sum of the α_i's and the β_i's, and the moving average part is characterized by minus the β_i coefficients. Such an analogy has been used by Bollerslev (1985b) to motivate the use of the autocorrelogram and partial autocorrelogram for ϵ_t^2 in model identification.

If ϵ_t^2 is generated by a GARCH(p,q) model, the forecast of the conditional variances is a non-trivial function of the information set today, just as in the ARCH(1) model illustrated in Section II. This allows a solution to the rational expectations models of the preceeding section. The conditional variance s steps ahead can be written:

$$(20) \qquad h_{t+s} = \omega + \sum_{i=1}^{n} [\alpha_i \epsilon^2_{t+s-i} + \beta_i h_{t+s-i}] + \sum_{i=s}^{m} [\alpha_i \epsilon^2_{t+s-i} + \beta_i h_{t+s-i}]$$

where $n = \min\{m, s-1\}$ and by definition summation from 1 to 0 and from $s > m$ to m both are equal to zero. Thus,

$$(21) \qquad E_t(h_{t+s}) = \omega + \sum_{i=1}^{n} [(\alpha_i + \beta_i) E_t(h_{t+s-i})] + \sum_{i=s}^{m} [\alpha_i \epsilon^2_{t+s-i} + \beta_i h_{t+s-i}]$$

which gives a recursive solution for the conditional expectations. In particular for the GARCH(1,1) model and $s \geq 2$ this expression reduces to

$$(22) \qquad E_t(h_{t+s}) = \omega + (\alpha + \beta)\, E_t(h_{t+s-1}).$$

In the stationary case where all the roots of the autoregressive polynomial $\alpha(z)+\beta(z)=0$ lie outside the unit circle and therefore the sum of the α_i's and the β_i's is less than one, the unconditional variance of ϵ_t is given by:

$$E(\epsilon^2_t) = \sigma^2 = \omega[1 - \sum_{i=1}^{m} (\alpha_i + \beta_i)]^{-1} .$$

By simple substitution equation (21) becomes

$$(23) \qquad E_t(h_{t+s}) = \sigma^2 + \sum_{i=1}^{n} [(\alpha_i + \beta_i)(E_t(h_{t+s-i}) - \sigma^2)] + \sum_{i=s}^{m} [\alpha_i \epsilon^2_{t+s-i} + \beta_i h_{t+s-i}]$$

and it follows that $E_t(h_{t+s}) \to \sigma^2$ as $s \to \infty$ when the process is stationary.

To illustrate the ideas discussed above consider the time series of weekly data on the exchange rate between the U.S. dollar and the Swiss franc from July 1973 through August 1985 for a total of 632 observations. The data are interbank closing spot bid prices on Wednesdays taken from the International Monetary Markets Yearbook. This particular series is chosen for the analysis because the first differences of the logarithm show no significant serial correlation in the mean, a finding also reported in the paper by Diebold and Nerlove (1985), who kindly provided the data. Thus, attention can simply be focused upon the variance models.

A plot of the first differences of the logarithm of the series is given in Figure 1. It follows, that even though the series seems to be uncorrelated over time, the observations are clearly not independent. There is a tendency for large changes to be followed by large changes but of unpredictable sign. The modified Box-Pierce test statistic for up to tenth order serial correlation takes the value Q(10) = 9.127 corresponding to the .480

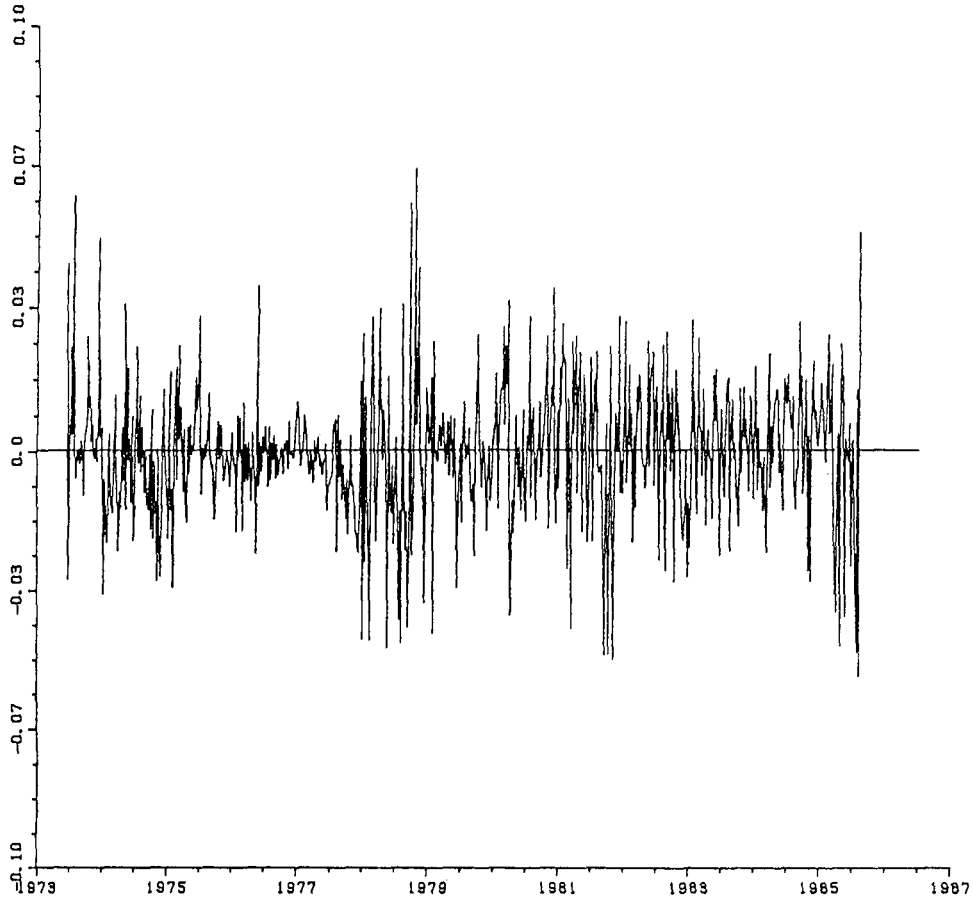


FIGURE 1

First Differences of Logarithm of U.S. Dollar/Swiss Franc Exchange Rate

fractile in a $\chi^2(10)$ distribution, whereas the test statistic for the first ten squared changes equals $Q^2(10) = 99.435$ which is highly significant at any level. For a precise definition of the Box-Pierce test statistic and a discussion of its applicability in

testing for absence of serial correlation in squared residuals see McLeod and Li (1983). The first ten autocorrelations, ρ_k, and partial autocorrelations, ϕ_{kk}, for the squared differences are given in Table I.

TABLE I

Correlation Structure for Squared Log Differences

k	1	2	3	4	5	6	7	8	9	10
ρ_k	.114	.117	.169	.212	.083	.110	.162	.012	.102	-.003
ϕ_{kk}	.114	.106	.148	.178	.024	.049	.094	-.070	.053	-.077

Judged by their asymptotic standard error, .040, both the autocorrelations and the partial autocorrelations die out fairly slowly. Thus, taking into account the small sample variability of the correlation estimates, cf. Bollerslev (1985b), a GARCH(1,1) model with $\alpha + \beta$ close to one seems a reasonable first model candidate.

Assuming conditional normality maximum likelihood estimation of the GARCH(1,1) model yields:[2]

$$\log(y_{t+1}/y_t) = \epsilon_{t+1}$$

$$\epsilon_{t+1}|\psi_t \sim N(0,h_{t+1})$$

[2]To start up the recursions defined by (7) and (18) we need presample values for ϵ_0^2 and h_0. A natural choice is given by the sample analogue $\frac{1}{T}\sum \epsilon_t^2$.

(24)

$$h_{t+1} = \underset{(1.36\cdot10^{-5})}{3.27\cdot10^{-5}} + \underset{(.011)}{.996\epsilon_t^2} \underset{(.011)}{-.871(\epsilon_t^2 - h_t)}$$

$$\log L = 1750.061$$

where the estimates in parentheses are asympototic standard errors obtained from the last auxilary regression in the Berndt, Hall, Hall and Hausman (1974) algorithm used to maximize the likelihood function.[3] Note, the estimated model in (24) does not have a finite unconditional fourth moment which may undermined the validity of the correlation structure for ϵ_t^2 as an identification tool. It is clear from the parameterization in (24) that the sum of α and β is very close to one, and that a conventional Wald type test would accept the hypothesis $\alpha + \beta = 1$. Multi-step forecast from (24) would approach the unconditional variance quite slowly as powers of .996 approach zero. The estimated mean lag of this variance expression, $1/(1-\beta)$, equals 7.752 or about eight weeks.

A series of Lagrange Multiplier tests for higher order ARCH, or equivalently GARCH, and moving average terms in ϵ_t were applied to determine the adequacy of the model specification. None of these tests detects model failure, which is quite surprising with over 600 observations. For instance the test for up to fourth order ARCH in addition to the GARCH(1,1) model equals .568, or .096 in a $\chi^2(3)$ distribution. The LM test for a constant in the

[3]Due to the complexity of the analytical derivatives for the GARCH model, numerical derivatives were used here and in what follows.

mean, i.e. a random walk with drift, equals .196, and the test for up to fourth order serial correlation takes the value 3.628, none of which are significant at traditional levels:

VI. INTEGRATED VARIANCE MODELS

The estimate for the GARCH(1,1) model as well as the theoretical expression in (23) suggests consideration of a particular class of GARCH models which have the property that the multi-step forecasts of the variance do not approach the unconditional variance. A GARCH model with this property is here defined to be integrated in variance.

DEFINITION

The GARCH(p,q) process characterized by the first two conditional moments:

$$E_{t-1}(\epsilon_t) = 0$$

$$E_{t-1}(\epsilon_t^2) = h_t = \omega + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}$$

where $\omega \geq 0$, $\alpha_i \geq 0$ and $\beta_i \geq 0$ for all i and the polynomial

$$1 - \sum_{i=1}^{q} \alpha_i z^i - \sum_{i=1}^{p} \beta_i z^i = 0$$

has d > 0 unit root(s) and max{p,q}-d root(s) outside the unit circle is said to be:

(i) Integrated in variance of order d if $\omega = 0$.

(ii) Integrated in variance of order d with trend if $\omega > 0$.

It is obvious, that for the GARCH(p,q) process to be integrated in variance a necessary condition is that the α_i's and the β_i's sum to one. In such models the multistep variance conditional on ψ_t will always depend upon h_{t+1}, but for shorter forecast horizons additional information will also be important in forming optimal forecasts of the conditional variances. The integrated GARCH(p,q) models, both with or without trend, are therefore part of a wider class of models with a property called "persistent variance" in which the current information remains important for the forecasts of the conditional variances for all horizons.

To illustrate consider the integrated GARCH(1,1) model

$$h_{t+1} = \alpha\epsilon_t^2 + (1 - \alpha)h_t$$

where $0 < \alpha < 1$. From (22) it follows that for this particular model

$$E_t(h_{t+s}) = h_{t+1} \quad . \tag{25}$$

The conditional variance s steps in the future is the same as the conditional variance one step ahead for all horizons s. This model is obviously very closely related to the traditional random walk with a unit root in the conditional mean. In a random walk without a drift the prediction of the mean s steps in the future is just equal to the level today. Thus, the information today is

important in forecasting forever. Shocks to to the system are permanent. In the same way shocks to the conditional variance in the integrated GARCH(1,1) model will not be forgotten.

In the integrated GARCH(1,1) model with trend

$$h_{t+1} = \omega + \alpha\epsilon_t^2 + (1 - \alpha)h_t$$

and therefore

$$E_t(h_{t+s}) = s\omega + h_{t+1} \ .$$

Again this model resembles very closely the traditional random walk with a drift in the mean. Even though the relative importance of h_{t+1} in the expression for the conditional variance s steps ahead decreases with the horizon, the effect is persistent.

The empirical plausibility of integrated GARCH models has already been established by the findings in Engle, Lilien and Robins (1985) and Bollerslev, Engle and Wooldridge (1985) that ARCH and GARCH models for interest rates typically exhibit parameters which are not in the stationary region. The estimates presented above for the U.S. dollar/Swiss franc exchange rate also suggests an integrated variance model.

Reestimating the GARCH(1,1) model imposing the restriction for integration in variance, yields:

$$(26) \qquad h_{t+1} = \underset{(.008)}{.090}\epsilon_t^2 + (1-.090)\, h_t$$

$$\log L = 1745.225.$$

The log likelihood is reduced below that of the unrestricted GARCH(1,1) by 4.836 which implies a likelihood ratio test statistic of 9.672 with two degrees of freedom. However, this reduction is almost entirely due to the restriction $\omega = 0$ since the t-statistic on this parameter equals 2.408. Indeed if the model is reestimated allowing a trend in the variance, the log likelihood is essentially unchanged relative to (23) and the parameter α is estimated to .118(.013). Here we leave out the intercept on the theoretical ground that a trend in the variance seems unreasonable for this particular model.

It follows from (19) that the integrated GARCH(1,1) model may be rewritten as

$$\begin{aligned} \Delta\epsilon_t^2 &= -(1-\alpha)(\epsilon_{t-1}^2 - h_{t-1}) + (\epsilon_t^2 - h_t) \\ &= -(1-\alpha)\,v_{t-1} + v_t \end{aligned}$$

Therefore, even though the population autocorrelation and partial autocorrelations for $\Delta\epsilon_t^2$ don't exist if (26) is the true model, it is interesting to note that the sample correlation structure for $\Delta\epsilon_t^2$ resembles that of an MA(1) process with moving average parameter close to minus one fairly well, see Table II.

The appropriate procedure for testing for integration in variance is not yet clear. It is possible that all of the well known difficulties when testing for unit roots in the mean as discussed by Dickey and Fuller (1979) applies in this context as

TABLE II

Correlation Structure for Differences of Squared Log Differences

k	1	2	3	4	5	6	7	8	9	10
ρ_k	-.514	.002	-.018	.113	-.100	.007	.110	-.133	.114	-.142
ϕ_{kk}	-.514	-.357	-.312	-.105	-.123	.129	.061	-.049	.090	-.079

well, see Engle and Granger (1985) for some recent references on this subject. However in a few simple Monte Carlo runs, the standard test statistics appear to be relatively well behaved. The knife edge condition for the unit root in variance does not seem as severe as in the mean. Estimation by maximum likelihood essentially does a GLS correction and this may be sufficient to give standard asymptotic normality. At the moment, this is a conjecture.

Another complication when testing in (26) and in GARCH models in general involve the issues of one-sided tests, as the null hypothesis is on the boundary of the parameter space. Therefore, under the above conjecture, it may be simpler to construct Lagrange Multiplier tests for the adequacy of model (26). The troublesome trend term, ω, then appears to be in the wrong tail as the likelihood function increases in the negative direction from zero. Thus, the local LM test will accept the null for any one sided test. Similarly, the LM test against the unrestricted GARCH(1,1) with $\omega = 0$ equals 3.565 which is close to the 95 percent fractile in a $\chi^2(1)$ distribution. However, the test is in the

explosive direction and therefore if one chooses to consider only stationary alternatives it would also accept the null at any level. The tests for other forms of ARCH and serial correlation in the mean are again insignificant. In particular the test for up to fourth order ARCH equals 4.026 corresponding to the .597 fractile in a $\chi^2(4)$ distribution, and the test for up to fourth order serial correlation takes the value 2.164 corresponding to the .294 fractile in a $\chi^2(4)$ distribution.

VII. DISTRIBUTION OF MULTISTEP FORECAST ERRORS AND TIME AGGREGATION

Section V examined the second conditional moment of multistep forecasts. In this section, the properties of higher order moments will be developed. Engle (1982) and Bollerslev (1985a) established that the unconditional distribution of ARCH and GARCH processes with conditionally normal errors in general have fatter tails than the normal, i.e. they are leptokurtotic. They also showed that the conditions for existence of finite unconditional fourth moments restrict the maximum temporal dependence by restricting the admissable parameter space, and therefore in particular that all integrated GARCH models will have infinite unconditional fourth moments in the population. In this section we discuss the distribution of the fourth conditional moments.

Although most of the applications of the ARCH and GARCH models have assumed that the conditional distribution is normal, there is no necessity for this assumption. Recently, Bollerslev (1985c) has postulated that the conditional distribution is a

standardized Student-t with unknown degrees of freedom υ. By estimating this parameter along with the other GARCH parameters it is possible not only to have a fat tailed unconditional distribution but also a fat tailed conditional distribution. If υ is twenty-five or larger the conditional distribution is essentially indistinguishable from the normal, while smaller values of υ give larger fourth conditional moments.

Consider now the GARCH(1,1) process characterized by the first four conditional moments:

$$
\begin{aligned}
&E_t(\epsilon_{t+1}) = 0 \\
&E_t(\epsilon^2_{t+1}) = h_{t+1} = \omega + \alpha\epsilon^2_t + \beta h_t \\
&E_t(\epsilon^3_{t+1}) = 0 \\
&E_t(\epsilon^4_{t+1}) = \kappa h^2_{t+1}.
\end{aligned}
\tag{27}
$$

Of course, for the GARCH(1,1) with conditionally normal errors $\kappa = 3$ and for the standardized Student t with $\upsilon > 4$ degree of freedom we have $\kappa = 3(\upsilon-2)(\upsilon-4)^{-1} > 3$ so the conditional distribution is also leptokurtic in that situation.

It follows that the fourth conditional moment of a forecast error s periods in the future equals

$$E_t(\epsilon^4_{t+s}) = \kappa E_t[E_{t+s-1}(h^2_{t+s})] = \kappa\, E_t(h^2_{t+s}),$$

and for $s \geq 2$

$$(28) \qquad E_t(h^2_{t+s}) = E_t[E_{t+s-2}(\omega + \alpha\epsilon^2_{t+s-1} + \beta h_{t+s-1})^2]$$

$$= E_t[\alpha^2(\kappa-1)h^2_{t+s-1} + (\omega + (\alpha+\beta)h_{t+s-1})^2] \ .$$

In the stationary case, the series defined by this expression will converge whenever the unconditional fourth moment exists, i.e. if $\kappa\alpha^2 + 2\alpha\beta + \beta^2 < 1$ see Bollerslev (1985a).

For the integrated GARCH(1,1) model where $\omega = 0$ and $\beta = 1-\alpha$, (28) reduces to

$$(29) \qquad E_t(h^2_{t+s}) = [1 + (\kappa-1)\alpha^2]^{s-1} \ h^2_{t+1} \ .$$

The further into the future this process is forecast, the larger the conditional kurtosis of the forecast errors. The rate of increase depends positively upon the kurtosis of the original process and upon the parameter α.

Of special interest for questions of time aggregation is the conditional kurtosis of the sum of future ϵ_t's. Define

$$(30) \qquad \kappa^{(2)}_t = E_t(\epsilon_{t+1} + \epsilon_{t+2})^4 \ / \ [E_t(\epsilon_{t+1} + \epsilon_{t+2})^2]^2$$

which will be the conditional kurtosis of a random walk process which is sampled every two periods. Therefore, if a conditional kurtosis of κ is found using weekly data, a kurtosis of $\kappa^{(2)}_t$ would be expected for biweekly data. Note, that in general the biweekly kurtosis will have a time subscript if the weekly model in (27) is true. Simple calculations yield:

$$E_t(\epsilon_{t+1}+\epsilon_{t+2})^4 = E_t(\epsilon_{t+1}^4) + E_t(\epsilon_{t+2}^4) + 6\,E_t(\epsilon_{t+1}^2\epsilon_{t+2}^2)$$

$$= \kappa\, h_{t+1}^2 + \kappa\, E_t(h_{t+2}^2) + 6\,E_t(\epsilon_{t+1}^2 h_{t+2}) \ .$$

Using (28) and substituting back into (30) we get:

$$(31) \qquad \kappa_t^{(2)} = \kappa + \kappa\,\frac{h_{t+1}^2[(3-\kappa)(\frac{2\beta}{\kappa} - \alpha^2) + 2\alpha(\alpha+2)] + 2\omega h_{t+1}(\frac{3-\kappa}{\kappa})}{[h_{t+1}(1+\alpha+\beta) + \omega]^2} \ .$$

Thus, the kurtosis of the conditional distribution for the two-step estimator will be smaller than the one-step kurtosis whenever the numerator in this expression is negative. In particular, when $3 < \kappa < 2\beta\alpha^{-2}$ it is possible to have a decrease in $\kappa_t^{(2)}$ relative to κ. For $\kappa = 3$ and $\alpha > 0$ there is an unambig- uous increase in kurtosis. If α and β are both zero, so the process defined in (27) has constant variance, $\kappa^{(2)}$ becomes time independent,

$$\kappa^{(2)} = \kappa + \frac{3-\kappa}{2}$$

and the familar result from the central limit theorem that $\kappa^{(2)}$ is closer to three than κ applies. The time dependence of $\kappa_t^{(2)}$ also vanishes for the integrated GARCH(1,1) model. In that situation the two step conditional kurtosis becomes:

$$(32) \qquad \kappa^{(2)} = \kappa + \kappa[(3-\kappa)(\frac{2(1-\alpha)}{\kappa} - \alpha^2) + 2\alpha(\alpha+2)] \ .$$

General conditions for a decrease in kurtosis is therefore easy to find in this case. For example if $\alpha = .1$, $\kappa^{(2)} < \kappa$ provided

$3.940 < \kappa < 137.060$, and for $\alpha = .05$ the condition for a decrease in kurtosis is $3.365 < \kappa < 677.635$.

For the exchange rate data, the integrated GARCH(1,1) model given in (26) was reestimated allowing for conditionally standardized Student-t distributed errors treating $1/\upsilon$ as an additional parameter to be estimated. The log likelihood function for a sample of T observations becomes in this situation

$$\text{(33)} \qquad \text{LogL} = \sum_{t=1}^{T} \left[\log(\Gamma(\tfrac{\upsilon+1}{2})) - \log(\Gamma(\tfrac{\upsilon}{2})) - \tfrac{1}{2}\log(\upsilon-2)h_t \right.$$
$$\left. - \tfrac{1}{2}(\upsilon+1)\ \log(1 + \epsilon_t^2 h_t^{-1}(\upsilon-2)^{-1}) \right] , \qquad \upsilon > 2$$

where $\Gamma(\cdot)$ denotes the usual gamma function. It is possible to show, that for $\upsilon \to \infty$, or equivalently $1/\upsilon \to 0$, (33) converges to the log likelihood function with conditionally normal errors defined in (7). Maximization by the BHHH algorithm leads to the estimates:

$$\epsilon_{t+1}|\psi_t \sim t_{\upsilon, h_{t+1}} , \qquad 1/\upsilon = \underset{(.009)}{.096}$$

$$\text{(34)} \qquad h_{t+1} = \underset{(.013)}{.104}\epsilon_t^2 + (1-.104)h_t$$

$$\log L = 1751.735 .$$

The likelihood ratio test statistic for conditional normality, i.e. $1/\upsilon = 0$, takes the value 13.019 which is highly significant at any reasonable level, see Bollerslev (1985c) where it is shown by means of Monte Carlo methods that the likelihood ratio test

statistic is more concentrated towards the origin than a $\chi^2(1)$ distribution in this situation. Also, a conventional t-test on $1/\upsilon$ overwhelmingly rejects the assumption of conditional normality for this model. It follows from the formula $\kappa = 3(\upsilon-2)(\upsilon-4)^{-1}$ that the conditional kurtosis in (34) is estimated to 3.94. This compares quite well with a sample kurtosis of the standardized residuals equal to 3.74. The unrestricted GARCH(1,1) was also estimated with conditionally Student-t distributed errors with the finding that $1/\upsilon = .115(.016)$. The likelihood ratio test statistic for $1/\upsilon = 0$ equals here 10.183, which again is highly significant. The estimated conditional kurtosis for this model is 4.28 while the sample analog is 3.93.

To examine the effects of time aggregation the series was split into two subseries of 315 non-overlapping biweekly observations. We confirm the findings by Boothe and Glassman (1985) and others summarized therein that the unconditional kurtosis declines with time aggregation dropping from 4.47 for the whole sample to an average of 3.55 for each of the two subsamples. However, if the true model is indeed the integrated GARCH(1,1) model in (34), the population fourth moment in each case is infinite. The decline in the sample kurtosis might therefore merely be due to the decline in sample size.

The interesting question in this context is the effect of time aggregation on the conditional fourth moment. Recognizing that an integrated GARCH(1,1) weekly model does not aggregate to an integrated GARCH(1,1) biweekly model, we proceed to estimate

this biweekly model. The result for the first subsample is

$$h_{t+1}^{(2)} = \underset{(.023)}{.118}\epsilon_t^2 + (1-.118)h_t^{(2)} , \qquad 1/\upsilon = \underset{(.002)}{.074} .$$

The estimated conditional kurtosis for this model is 3.63. This is in close accordance with the implied estimate from the weekly model (34). Substituting the parameter values from the estimated weekly model in (32) we obtain an estimate of $\kappa^{(2)} = 3.96$. Note, this estimate of $\kappa^{(2)}$ is essentially unchanged from the weekly estimate, $\kappa = 3.94$. The parameters are just at the point where the conditional kurtosis does not change with aggregation. The second biweekly sample has a larger estimated value of υ, and therefore a kurtosis closer to 3.

VIII. NON-LINEAR CONDITIONAL HETEROSKEDASTICITY

One explanation for the success of the conditional Student-t distribution in modelling the exchange rate series is that the observations are contaminated by some outliers or extreme values which make the conditional distribution look fat tailed. If indeed these extreme values are interpreted as outliers, then they may not be helpful in predicting future variances, and furthermore the estimates in the variance function may be unduely influenced by a few extreme observations. To examine this issue, a series of alternative specifications for the conditional variance were estimated. Here we shall only consider the two most successful of these alternative specifications. These are:

$$(35) \qquad h_{t+1} = \omega + \beta h_t + \gamma_1 |\epsilon_t|^{\lambda_1}$$

and

$$(36) \qquad h_{t+1} = \omega + \beta h_t + \gamma_2 (2F(\epsilon_t^2/\lambda_2) - 1)$$

where F denotes the standard normal cumulative distribution function. If $\lambda_1 = 2$ in (35) the model is simply GARCH(1,1). In (36) if λ_2 is large relative to the standard deviation of ϵ_t, the model is essentially linear.

Models (35) and (36) were both estimated with the same maximum likelihood routine using the BHHH algorithm simply by substituting in the new expressions for the conditional variances. Assuming conditional normality the likelihood function increases from 1750.061 for the unrestricted GARCH(1,1) to 1752.571 and 1753.461 for models (35) and (36) respectively. The likelihood ratio test statistic for (35) versus the unrestricted GARCH(1,1) equals therefore 5.020 corresponding to the .975 fractile in a $\chi^2(1)$ distribution. The maximum likelihood estimate of λ_1 is 1.543(.130). The estimated value of λ_2 in (36) is $.765 \cdot 10^{-3}$ $(.253 \cdot 10^{-3})$. Since the unconditional residual variance equals $\sigma^2 = .268 \cdot 10^{-3}$, it follows that $(\hat{\lambda}_2)^{1/2} = 1.690 \cdot \sigma$, and there is an important attenuation of the largest residuals in (36). Also, (36) exhibits a slightly higher likelihood function than (35) and converged more easily.

The improvements for the non-linear models are somewhat greater when the conditional Student-t distribution is employed. In that case the likelihood function increases from 1756.497 for

the unrestricted GARCH(1,1) to 1760.297 and 1761.730 for models (35) and (36) respectively. However, the non-linear models with conditionally t-distributed errors were much harder to iterate to convergence. Both the estimates of $\hat{\lambda}_1 = 1.428(.149)$ and $\hat{\lambda}_2 = .408 \cdot 10^{-3}(.148 \cdot 10^{-3})$ are lower when $1/\upsilon > 0$. It is interesting to note that the estimates of β are very similar across the six models, ranging from a high of .891(.013) for (36) with conditionally normal errors to a low of .852(.031) for (35) with conditionally t-distributed errors.

This new non-linear class of conditional variance models may provide important improvements for some time series, however in this case it is not clear that the improvements are worth the analytical complexity which arises from the non-linearities. An alternative approach to the non-linear models might be to use non-parametric techniques. For a recent discussion and some references on this subject see Engle, Granger, Rice and Weiss (1986).

IX. SOLVING THE ASSET PRICING EQUATION

Using the class of models described above it is now possible to obtain simple closed form solutions to the asset pricing equations given in (10), (15) and (16). In each case a solution is a stochastic process for the asset price which depends upon the process of the forcing variable. In particular, for the forward contract in (10) the process of the price should depend upon the process of the spot price, y_t,

$$(10) \qquad p_t = r^{-s}[E_t(y_{t+s}) - \delta V_t(y_{t+s})].$$

There are several special cases to be considered in solving equation (10). To illustrate let y_t be a random walk with innovations which follow an integrated GARCH(1,1) process. In that situation $E_t(y_{t+s}) = y_t$ and

$$V_t(y_{t+s}) = E_t(\sum_{i=1}^{s} \epsilon_{t+i}^2) = E_t(\sum_{i=1}^{s} h_{t+i}) = sh_{t+1}$$

so that

$$(37) \qquad p_t = r^{-s}[y_t - \delta sh_{t+1}] .$$

For a future contract where no money changes hands until the terminal date at t+s, the alternative sure rate of return is equal to zero so that r=1. Thus, in that case (37) simplifies to:

$$(38) \qquad p_t = y_t - \delta sh_{t+1} .$$

Therefore, if $\delta \neq 0$ there will be a time varying risk premium in the future contract. For contracts far in the future new information will have a substantial effect on asset prices as it changes agent's perceptions of the variance of the final payoff as well as all the intermediate variances. The persistence of the variance gives time varying risk premia even for contracts many periods in the future and implies sizeable effects on asset prices. Note, because the innovation in y_t enters both linearly and as a square in (37) the future price series may exhibit some skewness.

Suppose now as an alternative case, that y_t is a simple random walk with serially independent innovations and variance σ^2. Then $V_t(y_{t+s}) = s\sigma^2$ and the solution to (10) proceeds in identical fashion, leading to the future contract pricing equation:

$$p_t = y_t - \delta s\sigma^2 \quad . \tag{39}$$

This is essentially the solution achieved by Hellwig (1982), although in a different context. The variance of the spot price also enters the pricing equation in this situation, but does not give rise to a time varying risk premium. New information casts no light on future uncertainty.

Finally, consider the case where the spot price y_t is a random walk with innovations distributed as a stationary GARCH(1,1). Letting $\sigma^2 = \omega/(1-\alpha-\beta)$, it follows from (23) that the multi-step forecasts of the conditional variance of the innovations, h_{t+s}, can be written as

$$E_t(h_{t+s} - \sigma^2) = (\alpha+\beta)^{s-1}(h_{t+1} - \sigma^2)$$

and

$$V_t(y_{t+s}) = \sum_{i=1}^{s} \sigma^2 + E_t(h_{t+i} - \sigma^2)$$

$$= s\sigma^2 + [h_{t+1} - \sigma^2] \frac{1 - (\alpha+\beta)^s}{1 - \alpha - \beta} \quad .$$

By substitution in (10) the solution for the future constract is:

$$(40) \qquad p_t = y_t - \delta s\sigma^2 + \delta[h_{t+1} - \sigma^2] \frac{1 - (\alpha+\beta)^s}{1 - \alpha - \beta} .$$

When $\alpha+\beta < 1$, the current information continues to be an important part of the time varying risk premium even for large s, but the relative importance decreases with the length of the contract. This is in contrast to the solution for the integrated GARCH(1,1) model given in (38) where the importance of h_{t+1} increases with the length of the contract.

Of course, the set of solutions presented here is far from exhaustive, but merely serves to illustrate the general idea. Namely, that a given solution to the asset pricing equations (10), (15) or (16) depends in a crucial way on the distribution of the forcing variable, y_t, and in particular on the conditional variance of y_t.

X. CONCLUSIONS

The statistical properties as well as the empirical relevance of the ARCH model in modelling risk and uncertainty in economies is by now well established. In this paper, however, several new unanswered questions have been raised.

Both from an empirical and theoretical economic point of view do the integrated GARCH models constitute an interesting development, but the statistical properties for this new class of models are largely unknown. Some preleminary Monte Carlo evidence suggests that the knife edge properties when estimating and testing for a unit root in the mean might not be as severe in the inte-

grated variance models. So far this is a conjecture. Similarly, the empirical regularity and statistical properties of the more complicated non-linear conditonal heteroskedasticity models remain open questions. Another related issue involves the problem of testing for one sided alternatives when the null hypothesis is on the boundary of the admissable parameter space, as is typically the case when testing in GARCH models with non-negativity constraints. It would also be of interest to see how the aggregation formulas derived here hold up against a more detailed and broader empirical analysis. Finally, the validity of the asset pricing formulas presented throughout the paper, especially the pricing equations for futures contracts given in the last section, in explaining observed price behavior is another very interesting question. We leave the answer to all of these questions for future research.

ACKNOWLEDGEMENTS

We are deeply indebted to many for discussions helping to clarify key issues. Particular thanks go to Dick Baillie, Buz Brock, Roman Frydman, Clive Granger, Dale Poirier, Mike Rothschild, Larry Weiss and Mike Wickens. For remaining misunderstandings omissions and errors, we humbly apologize. The research was supported by NSF Grant No. SES 84-20680.

REFERENCES

Amsler, C., (1984a). Term Structure Variance Bounds and Time Varying Liquidity Premia. Economics Letters, 16, 137-144.

_____ (1984b). Including Time Varying Liquidity Premia in Term Structure Variance Bounds. Michigan State University: Econometrics Workshop Paper #8308.

Berndt, E.K., Hall, B.H., Hall, R.E., & Hausman J.A., (1974). Estimation and Inference in Nonlinear Structural Models. Annals of Economic and Social Measurement, 4, 653-665.

Blanchard, O.J., & Watson, M.W., (1984). Are Business Cycles All Alike? NBER Working Paper No. 1392.

Bodie, Z., A. Kane, & McDonald, R., (1984). Why Haven't Nominal Rates Declined. Financial Analysts Journal, March-April 1984.

_____ (1984). Why are Real Interest Rates So High? NBER Working Paper No. 1141.

Bollerslev, T., (1985a). Generalized Autoregressive Conditional Heteroskedasticity. Forthcoming Journal of Econometrics. Also UCSD, Department of Economics Discussion Paper No. 85-16.

_____ (1985b). A Note on the Correlation Structure for the Generalized Autoregressive Conditional Heteroskedastic Process. UCSD, Department of Economics, Discussion Paper No. 85-23.

_____ (1985c). A Conditionally Heteroskedastic Time Series Model for Security Prices and Rates of Return Data. UCSD, Department of Economics, Discussion Paper No. 85-32.

Bollerslev, T., Engle, R.F., & Wooldridge, J.M., (1985). A Capital Asset Pricing Model with Time Varying Covariances. UCSD, Department of Economics, Discussion Paper No. 85-28.

Boothe, P., & Glassman, P., (1985). The Statistical Distribution of Exchange Rates: Empirical Evidence and Economic Implications. University of Alberta, Department of Economics, Research Paper No. 85-22.

Coulson, N.E., & Robins, R.P., (1985). Aggregate Economic Activity and the Variance of Inflation: Another Look. Economics Letters, 17, 71-75.

Dickey, D.A., & Fuller, W.A., (1979). Distribution of Estimators for Autoregressive Time Series with a Unit Roof. Journal of the American Statistical Association, 77, 427-431.

Diebold, F.S., & Nerlove, N., (1985). ARCH Models of Exchange Rate Fluctuations. Unpublished manuscript, University of Pennsylvania, Department of Economics.

Diebold, F.S., & Pauly, P., (1985). Endogenous Risk in a Portfolio Balance Rational Expectations Model of the Deutschemark-Dollar Rate. University of Pennsylvania, Department of Economics, Econometric Discussion Paper # 85-17.

Domowitz, I., & Hakkio, C.S., (1985). Conditional Variance and the Risk Premium in the Foreign Exchange Market. Journal of International Economics, 19, 47-66.

Engle, R.F., (1982). Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of U.K. Inflation. Econometrica, 50, 987-1008.

_____ (1983). Estimates of the Variance of U.S. Inflation Based on the ARCH Model. Journal of Money Credit and Banking, 15, 286-301.

Engle, R.F., Brown, S., & Stern, G., (1984). Short Run Forecasts of Electricity Sales: A Comparison of New Methodologies. Unpublished manuscript UCSD, Department of Economics.

Engle R.F., & Granger, C.W.J., (1985). Cointegration and Error Correction: Representation, Estimation and Testing. Forthcoming Econometrica. Also manuscript UCSD, Department of Economics.

Engle, R.F., Granger, C.W.J, & Kraft, D., (1984). Combining Competing Forecasts of Inflation Using a Bivariate ARCH Model. Journal of Economic Dynamics and Control, 8, 151-165.

Engle, R.F., Granger, C.W.J., & Rice, J., Weiss, A., (1986). Non-Parametric Estimates of the Relation Between Weather and Electricity Demand. Forthcoming Journal of the American Statistical Association, June 1986.

Engle, R.F., Hendry, D.F. & Trumble, D., (1985). Small-sample Properties of ARCH Estimators and Tests. Canadian Journal of Economics, 18, 66-93.

Engle, R.F., Hendry, D., & Richard, R.F., (1983). Exogeneity. Econometrica, 51, 277-304.

Engle, R.F., & Kraft, D., (1983). Multiperiod Forecast Error Variances of Inflation Estimated from ARCH Models. Applied Time Series Analysis of Economic Data, (A. Zellner, Ed.), Bureau of the Census, 293-302.

Engle, R.F., Lilien, D., & Robins, R., (1985). Estimating Time Varying Risk Premia in the Term Structure: The ARCH-M Model. Forthcoming Econometrica. Also UCSD, Department of Economics, Discussion Paper No. 85-17.

Engle, R.F., & Watson, M., (1985). Applications of Kalman Filtering in Econometrics. Invited Paper to the Fifth World Congress of the Econometric Society, Cambridge, Mass. Forthcoming in Volume of Proceedings. Also UCSD, Department of Economics, Discussion Paper No. 85-31.

French, K.R., Schwert, G.W., & Stambaugh, R.F., (1985). Expected Stock Returns and Volatility. University of Rochester, Graduate School of Management, Working Paper Series No. MERC 85-10.

Granger, C.W.J., Robins, R.P., & Engle, R.F., (1984). Wholesale and Retail Prices: Bivariate Time Series Modelling with Forecastable Error Variances. In Model Reliability, (Belesley and Kuh, Eds.), MIT Press.

Hellwig, M.F., (1982). Rational Expectations Equilibrium With a Conditioning on Past Prices: A Mean Variance Example. Journal of Economic Theory, 26, 279-312.

Hsieh, D.A., (1985). The Statisitcal Properties of Daily Foreign Exchange Rates: 1974-1983. Mimeo, University of Chicago, Graduate School of Business.

Kraft, D., & Engle, R.F., (1982). Autoregressive Conditional Heteroskedasticity in Multiple Time Series Models. UCSD, Department of Economics, Discussion Paper No. 82-23.

McCurdy, T.H., & Morgan, I.G., (1985). Testing the Martingale Hypothesis in the Deutschmark/U.S. Dollar Futures and Spot Markets. Queens University, Department of Economics, Discussion Paper No. 639.

McLeod, A.J., & Li, W.K., (1983). Diagnostic Checking ARMA Time Series Models Using Squared-Residual Autocorrelations. Journal of Time Series Analysis, 4, 269-273.

Milhøj, A., (1985a). The Moment Structure of ARCH Processes. Scandinavian Journal of Statistics, 12, 281-292.

_____ (1985b). A Conditional Variance Model for Daily Deviations of an Exchange Rate. Unpublished manuscript, University of Copenhagen, Institute of Statistics.

Nemec, A.F., Linnell, (1984a). Conditionally Heteroscedastic Autoregressions. University of Washington, Department of Statistics, Technical Report #43.

_____ (1984b). Least Squares Estimation of Conditionally Heteroscedastic Autoregressions. University of Washington, Department of Statistics, Technical Report #48.

Pagan, A.R., Hall, A.D., & Trivedi, P.K., (1983). Assessing the Variability of Inflation. Review of Economic Studies, 50, 585-596.

Pantula, S.G., (1984). Autoregressive Conditionally Heteroscedastic Models. Unpublished manuscript, North Carolina State University, Department of Statistics.

Poterba, M.P., & Summers, L.H., (1984). The Persistance of Volatility and Stock Market Fluctuations. Harvard University, Institute of Economic Research, Discussion Paper No. 1092.

Wallis, K.F., (1980). Econometric Implications of the Rational Expectations Hypothesis. Econometrica, 48, 49-73.

Weiss, A.A., (1982). Asymptotic Theory for ARCH Models: Stability, Estimation and Testing. Forthcoming Econometric Theory. Also UCSD, Department of Economics, Discussion Paper No. 82-36.

Weiss, A.A., (1984). ARMA Models with ARCH Errors. Journal of Time Series Analysis, 5, 129-143.

DATA APPENDIX

Weekly Swiss Franc/Dollar Exchange Rate

(data to be read across rows)

2.7470	2.8690	2.7820	2.7850
2.8530	2.8640	3.0390	3.0410
3.0210	3.0180	3.0190	3.0100
3.0170	3.0210	2.9890	3.0000
3.0190	3.1000	3.1320	3.1710
3.1780	3.1970	3.1960	3.1900
3.1990	3.1910	3.3520	3.3640
3.3740	3.3910	3.2760	3.2540
3.1960	3.1260	3.1220	3.1340
3.0950	3.0490	3.0260	3.0650
3.0730	3.0390	2.9670	2.9380
2.8990	2.9150	2.8940	2.9940
2.9520	3.0090	3.0020	3.0100
2.9730	2.9970	2.9330	2.8930
2.9630	2.9850	2.9760	3.0140
2.9990	3.0160	3.0150	2.9850
2.9850	2.9470	2.9050	2.9020
2.8470	2.8740	2.8140	2.8070
2.7700	2.6850	2.6860	2.6250

2.5470	2.5250	2.5610	2.5790
2.5700	2.5440	2.4910	2.5370
2.5010	2.4780	2.3980	2.4360
2.4840	2.4670	2.5280	2.5350
2.5610	2.5700	2.5450	2.5580
2.5200	2.4770	2.4910	2.4780
2.4930	2.4910	2.4940	2.4950
2.5380	2.5620	2.5950	2.6770
2.7060	2.6780	2.6800	2.6750
2.6790	2.6760	2.6870	2.7230
2.7190	2.7080	2.6980	2.6540
2.6260	2.6260	2.6440	2.6390
2.6550	2.6690	2.6500	2.6340
2.6280	2.6270	2.6210	2.5990
2.6000	2.6110	2.6020	2.5930
2.5430	2.5640	2.5610	2.5810
2.5790	2.5300	2.5580	2.5390
2.5430	2.5250	2.5300	2.5190
2.4890	2.5000	2.4920	2.4650
2.4050	2.4990	2.4780	2.4830
2.4710	2.4790	2.4780	2.4920
2.4960	2.4750	2.4890	2.4720
2.4770	2.4720	2.4810	2.4650
2.4690	2.4550	2.4480	2.4460
2.4480	2.4350	2.4390	2.4410
2.4470	2.4370	2.4540	2.4520
2.4520	2.4480	2.4450	2.4550
2.4780	2.5060	2.5250	2.5160
2.5150	2.5140	2.5350	2.5530
2.5650	2.5570	2.5430	2.5440
2.5370	2.5180	2.5240	2.5180
2.5200	2.5280	2.5190	2.5140
2.1020	2.1220	2.1720	2.1650
2.1540	2.1460	2.1390	2.0900
1.9900	1.9620	1.9750	1.8810
1.8530	1.8990	1.8860	1.7940
1.7610	1.7770	1.7810	1.7810
1.8380	1.8200	1.8160	1.7980
1.8140	1.8700	1.8560	1.8550
1.8930	1.8960	1.9130	1.8720
1.8790	1.8540	1.8820	1.9000
1.9340	1.9650	1.9650	1.9580
1.9510	1.9340	1.9320	1.9820
1.9750	2.0140	2.0520	2.1000
2.0830	2.1080	2.1400	2.1280
2.0730	2.0930	2.1030	2.1560
2.1180	2.0580	2.1160	2.1220
2.1510	2.1410	2.1730	2.2020
2.1340	2.1750	2.2040	2.2180
2.2250	2.2020	2.1610	2.1150
2.0850	2.0850	2.0230	1.9980
1.9520	1.9310	1.9910	2.0000

2.0280	2.0130	1.9830	2.0360
2.0490	2.0580	2.0460	2.0760
2.0860	2.0490	2.0540	2.0650
2.0620	2.0580	2.0250	2.0610
2.0760	2.1160	2.1330	2.1410
2.0880	2.1090	2.1230	2.1130
2.0900	2.1180	2.1540	2.1910
2.1390	2.1530	2.1790	2.1750
2.1790	2.1620	2.1320	2.0940
2.1250	2.1140	2.1230	2.1550
2.1620	2.1780	2.1780	2.1570
2.1850	2.2120	2.2140	2.1890
2.2330	2.2530	2.2320	2.2440
2.2350	2.2290	2.1970	2.1940
2.1680	2.1160	2.1280	2.1770
2.1580	2.1710	2.1650	2.1900
2.2160	2.2480	2.2800	2.2680
2.2770	2.2650	2.2330	2.2710
2.3060	2.3320	2.3740	2.4030
2.4210	2.4340	2.4650	2.4450
2.4110	2.4000	2.3910	2.4640
2.4900	2.5340	2.5080	2.4950
2.5370	2.5580	2.4850	2.4940
2.4170	2.4500	2.5020	2.5210
2.5450	2.5550	2.5580	2.5760
2.6170	2.6520	2.6760	2.6830
2.6760	2.7370	2.8120	2.8230
2.8400	2.8980	2.8500	2.7550
2.6500	2.6420	2.6530	2.5310
2.5950	2.6400	2.6860	2.5810
2.5890	2.5900	2.5840	2.6010
2.5300	2.5350	2.5420	2.4510
2.3350	2.3690	2.2440	2.3620
2.5000	2.4930	2.4890	2.4930
2.4580	2.4370	2.4180	2.4020
2.3880	2.4030	2.4180	2.4360
2.3780	2.3980	2.3830	2.3890
2.3610	2.3530	2.3350	2.2970
2.2820	2.2370	2.2190	2.2270
2.2080	2.1900	2.1600	2.1190
2.0760	2.0260	2.0180	2.0510
1.9600	2.0140	1.9610	1.9860
1.9650	1.9120	1.8270	1.8350
1.8930	1.9390	1.9160	1.8750
1.8620	1.8670	1.9300	1.9570
1.9530	1.9720	1.9870	1.9710
1.8800	1.9130	1.8930	1.8650
1.8590	1.8180	1.8070	1.8130
1.7840	1.7130	1.6890	1.6130
1.6690	1.6530	1.6140	1.5910
1.5250	1.4920	1.5810	1.5420
1.5130	1.5060	1.6090	1.6190

1.6490	1.7210	1.7230	1.7090
1.7020	1.6410	1.6200	1.6440
1.6640	1.6670	1.6740	1.7030
1.6300	1.6720	1.6740	1.6710
1.6670	1.6770	1.6870	1.6890
1.7040	1.7150	1.7100	1.7110
1.7230	1.7130	1.7270	1.7320
1.7210	1.7340	1.7310	1.6750
1.6520	1.6510	1.6540	1.6260
1.6380	1.6570	1.6540	1.6560
1.6550	1.6640	1.6480	1.6280
1.6150	1.5750	1.5790	1.6030
1.6470	1.6570	1.6520	1.6380
1.6420	1.6440	1.6340	1.6030
1.6010	1.6030	1.5840	1.5710
1.5750	1.5830	1.6120	1.6250
1.6030	1.6170	1.6410	1.6650
1.7140	1.7250	1.7680	1.7960
1.8600	1.7880	1.7200	1.7000
1.6670	1.6500	1.6640	1.6650
1.6510	1.6530	1.6300	1.6240
1.6400	1.6190	1.5920	1.5980
1.5990	1.6500	1.6320	1.6420
1.6590	1.6570	1.6300	1.6300
1.6310	1.6500	1.6400	1.6380
1.6440	1.6620	1.6930	1.7390
1.7070	1.7070	1.7330	1.7520
1.8200	1.8010	1.7700	1.7870
1.7710	1.8060	1.8260	1.8810
1.9210	1.9620	1.9740	1.9190
1.9430	1.9380	1.8570	1.9050
1.9130	1.9480	2.0010	1.9810
2.0160	2.0630	2.0820	2.0420
2.0790	2.1070	2.0970	2.0520
2.0270	2.0720	2.1120	2.0670